Lecture 7: Two Variables: Correlation and Causation

Lecturer

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CLASS CODE

TOAE203

COURSE

Statistics and Probability in Economics and Business

AGENDA

01

TWO CATEGORICAL VARIABLES

The contingency table of Session 4, read as a relationship rather than as a probability

02

THE CORRELATION COEFFICIENT

From the scatter diagram to r — and the four things r cannot see

03

PREDICTING ONE VARIABLE FROM ANOTHER

The fitted line, the slope in its correct units, and what the slope is not

04

INTERPRETING CAUSALITY

Three causal worlds that produce identical coefficients

05

SIMPSON'S PARADOX

An aggregate comparison that reverses inside every single group

06

FROM NUMBER TO CLAIM

The three conditions, and the difference between predicting and intervening

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TWO CATEGORICAL VARIABLES

Profitability by Industry, in Counts

Record an answer before proceeding

Industry	Profitable	Not profitable	Total
Manufacturing	___	___	___
Service	___	___	___
Finance	___	___	___
___	___	___	___
Total	___	___	___

Which industry has the highest share of profitable firms?

The Same Table as Row Percentages

The order changes

The industry holding the largest count of profitable firms is not the industry holding the largest share. Neither table is incorrect: the first reports how many, the second how often.

- Largest count: industry ___
- Largest share: industry ___, at ___%
- The two are different rows of the same table

Industry	Profit	Non-profit	Total
Real-estate			
Services			
Construction			
Power			

A count measures the size of the group before it measures anything else.

TWO CATEGORICAL VARIABLES

Counts and Conditional Percentages

THE MEASURE

Two directions of division

Dividing by the row total gives the distribution of profitability within each industry. Dividing by the column total gives the distribution of industry within each profitability class.

- By row: of firms in this industry, what share are profitable
- By column: of profitable firms, what share are in this industry

THE CLAIM

The direction is a decision

Groups of unequal size cannot be compared on counts. Which way to normalize is chosen by the analyst, not by the data, and it fixes which question the table answers.

- Session 4 stated this as $P(A|B) \neq P(B|A)$

Two Variables: Three Combinations of Type

The shape of this session

Pair	Summarised by	Relationship measured by
categorical × categorical	contingency table	comparison of the rows
measurement × measurement	scatter diagram	r, then a fitted line
categorical × measurement	group means	difference between means

TWO CATEGORICAL VARIABLES

The Conditional Distribution

Reading a whole row rather than a single cell

	Probability Concept	Today
Object read	one conditional probability	one whole row
Operation	divide a cell by a row total	compare the rows with one another
Question	How likely is profit in Retail	Does industry tell us anything about profitability

Today stays descriptive. Whether a difference is large enough to be real is Session 11.

TWO CATEGORICAL VARIABLES

Association and Independence

THE VISUAL TEST

If every row is the same, there is no association

Two tables of row percentages, same variables, different data. The first adds nothing when the industry is known; the second changes the estimate substantially.

- Table A — 61%, 60%, 62%: knowing the industry adds nothing
- Table B — 84%, 47%, 31%: knowing it changes a great deal
- Session 4 called the first case independence

TWO CATEGORICAL VARIABLES

Practice 1 — One Table, Read Three Ways

Excel · PivotTable · industry × profitable

Build the PivotTable of industry × profitable. Show it as counts, then % of Row Total, then % of Column Total.

TIME

10 minutes

QUANTITY

Three views of one table, plus two cells of arithmetic

CONSTRAINT

Write down one question that the row table answers and the column table does not.

Then check P(A ∩ B) against P(A) × P(B). The three views are item 1 of the submission.

The Strength of an Association, Read Coarsely

Spread across rows	Reading
2 to 3 percentage points	negligible
10 to 15 points	moderate
40 points or more	strong

A single number for this strength is Cramér's V — Discussed in later lecture

A Question Left Open

Does entering this industry make a firm more likely to be profitable?

Firms in industry ___ are profitable ___% of the time; firms in industry ___ are profitable ___% of the time. The difference is ___ percentage points. The question is recorded and not answered today.

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Why the Contingency Table Fails

	Categorical pair	Measurement pair
Distinct values	8 industries × 2 outcomes	___ ROA values × ___ debt values
Table size	16 cells	approximately ___ cells
Firms per cell	tens	zero or one

What if the question is not Profitability (Yes or No) but ROA (an exact number)

THE CORRELATION COEFFICIENT

Reading a Scatter Diagram

Four questions, asked in this order

Direction

Does the cloud rise or fall?

Form

Is it straight or curved? Asked before any coefficient is computed.

Strength

Do the points cluster tightly or scatter widely?

Outliers

Does any point stand apart from the rest?

Convention: the variable to be explained goes on the vertical axis. That is already a direction of thought.

Covariance (Hiệp phương sai)

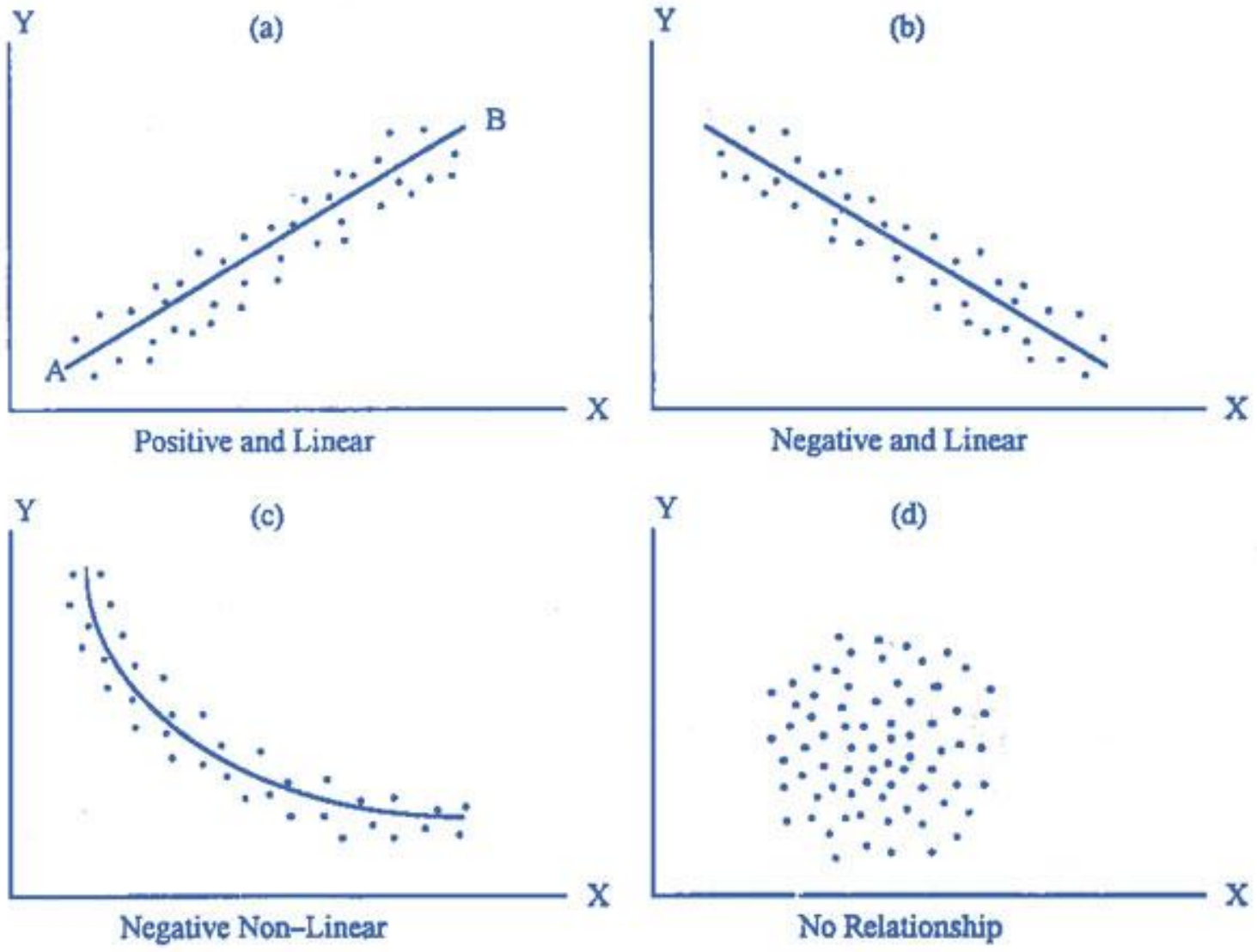
Population Covariance Formula

$$Cov(x,y) = \frac{\Sigma(x_i-\bar{x})(y_i-\bar{y})}{N}$$

Sample Covariance

$$Cov(x,y) = \frac{\Sigma(x_i-\bar{x})(y_i-y)}{N-1}$$

A cloud on one diagonal sums to one sign. A cloud spread across all four sums to near zero.



THE CORRELATION COEFFICIENT

Covariance: the Four Quadrants

$$\text{cov}(X,Y) = \Sigma\ (x_i - \bar{x})(y_i - \bar{y}) \,/\, (n - 1)$$

+

Upper right

Above average on both. Positive times positive.

+

Lower left

Below average on both. Negative times negative.

−

Upper left

Below on X, above on Y. Negative product.

−

Lower right

Above on X, below on Y. Negative product.

THE CORRELATION COEFFICIENT

Covariance Depends on the Units

———

cov in billions of VND

The relationship between the two variables

$\times 10^9$

the same cov, in VND

The same relationship, unchanged in every respect

The sign of a covariance is informative. Its magnitude is not.

The Pearson Correlation Coefficient

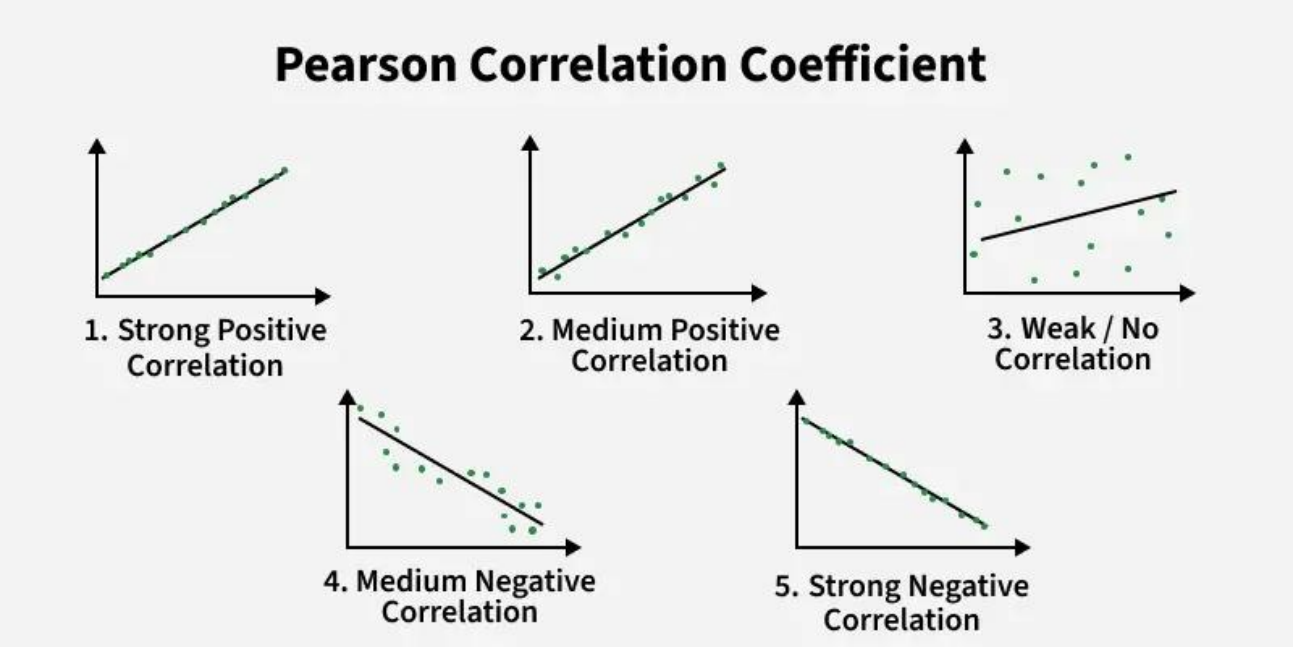
Covariance computed on standardized variables

NOT A NEW CONSTRUCTION

r = cov(X,Y) / (s_x · s_y) = cov(z_X , z_Y)

A statistical measure that expresses the strength and direction of a **linear** relationship between two variables

- Bounded: −1 ≤ r ≤ 1
- Unit-free: changing units or origin leaves r unchanged
- r(X,Y) = r(Y,X)



$$r = \frac{1}{n-1}\sum_{i=1}^{n}\left(\frac{x_i - \bar{x}}{S_x}\right)\left(\frac{y_i - \bar{y}}{S_y}\right)$$

The Pearson Correlation Coefficient

Covariance computed on standardized variables

$$r = \frac{S_{xy}}{\sqrt{S_{xx}}\sqrt{S_{yy}}}$$

where

$$S_{xy} = \sum (x - \bar{x})(y - \bar{y})$$

$$S_{xx} = \sum (x - \bar{x})^2, \; S_{yy} = \sum (y - \bar{y})^2$$

Size of Correlation	Interpretation
.90 to 1.00 (−.90 to −1.00)	Very high positive (negative) correlation
.70 to .90 (−.70 to −.90)	High positive (negative) correlation
.50 to .70 (−.50 to −.70)	Moderate positive (negative) correlation
.30 to .50 (−.30 to −.50)	Low positive (negative) correlation
.00 to .30 (.00 to −.30)	negligible correlation

The Third Property of r

$$r(X,Y) = r(Y,X)$$

The correlation of X with Y is identical to the correlation of Y with X. No information about which variable acts on which is present in the number. This is not a limitation awaiting a better formula — it is a property of the definition.

The number is symmetric. However, the question almost never is symmetric

Four Things r Does Not See

01

Non-linear form

A parabola: r ≈ 0 while the relationship is exact. Session 3, Anscombe.

02

A single outlier

Nineteen unrelated points and one distant point give r = 0.8.

03

The slope

Two clouds with the same r = 0.9 whose slopes differ fivefold.

04

Two groups combined

Two clouds, each falling, whose union rises.

The fourth is the contingency table, drawn as a picture.

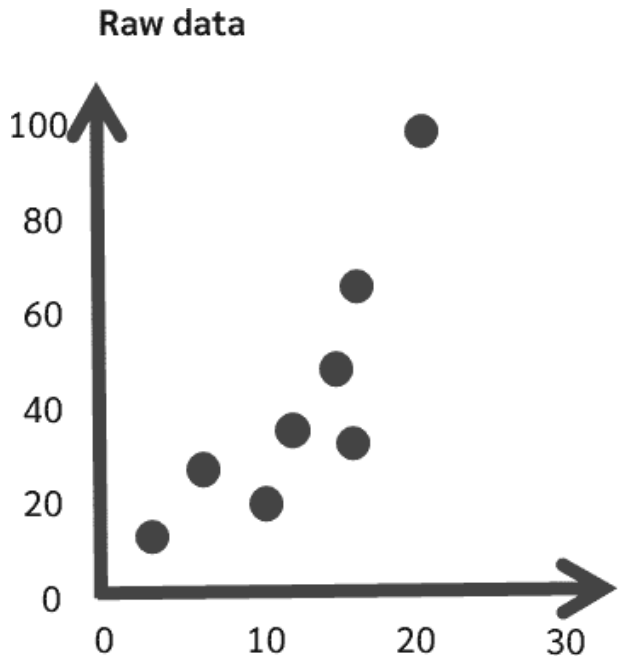
The Spearman Correlation Coefficient

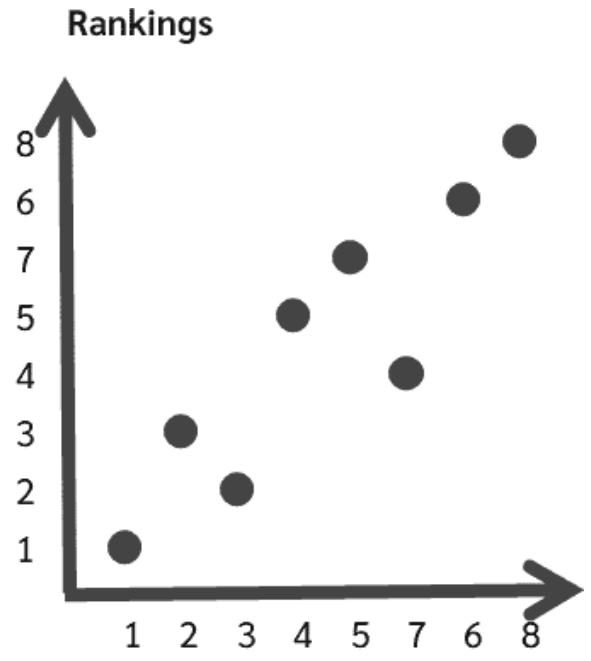
The Pearson Correlation applied to ranking

BUILT FROM THE FORMULA OF PEARSON

A statistical test that measures the strength and direction of a **monotonic** relationship between two variables

- Monotonic relationship means that as one variable increases, the other variable also tends to increase
- Spearman's correlation uses the rank order of the data instead of the actual raw values, sorts each variable from lowest to highest and assigns numbers (1, 2, 3, etc.) based on their positions.
- It then runs a standard Pearson correlation calculation on those ranked values.





$$r_s = 1 - \frac{6 \cdot \sum d_i^2}{n \cdot (n^2 - 1)}$$

Pearson and Spearman

Spearman is Pearson computed on the ranks

Pair	Pearson	Spearman
revenue × total assets	___	___
log(revenue) × log(total assets)	___	___
Use Pearson	straight form, no dominant observation	
Use Spearman	outliers, strong skew, or ordinal data	

Ranking repairs two of the four blindnesses: the outlier, and the monotone curve.

Practice 2 — The Correlation Matrix

Excel · Data Analysis > Correlation · conditional formatting

Build the correlation matrix of the seven principal measurement variables and apply a colour scale.

TIME

5 minutes

QUANTITY

Seven variables — 21 distinct coefficients

CONSTRAINT

Do not yet interpret any of them. Write down only which pair is highest.

The matrix is item 2 of the submission, with one sentence on whether the highest pair is surprising.

Spurious Correlation (Tương quan giả)

CIRCUMSTANCE 1

Pure Coincidence. Among 21 coefficients a high value arises by chance alone; among 200 it is a certainty.

CIRCUMSTANCE 2

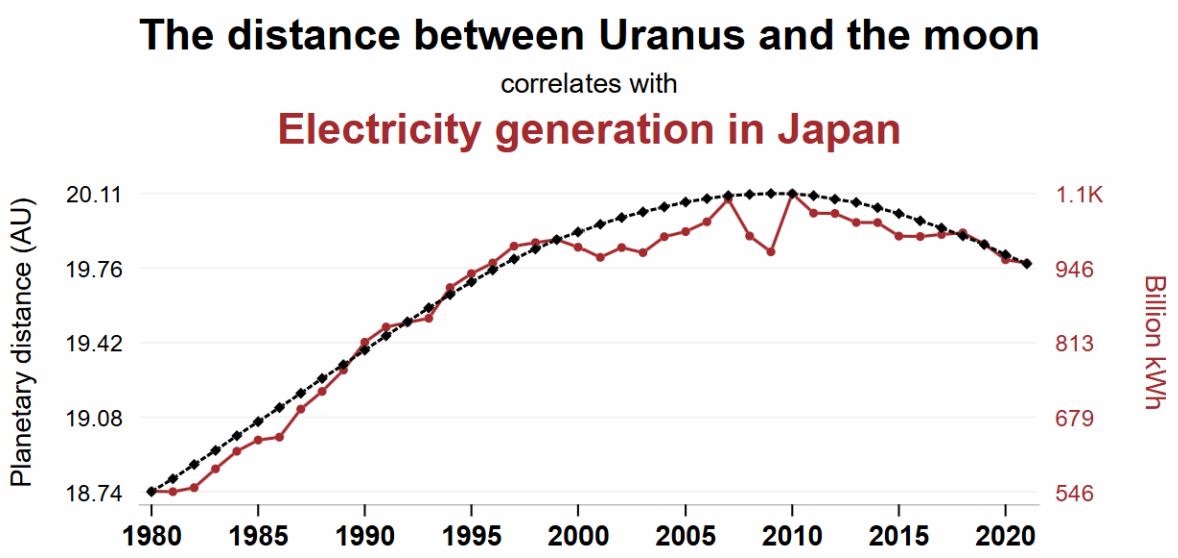
A shared time trend. Any two series that both grow over time correlate strongly, related or not — Session 14 calls this a spurious regression.

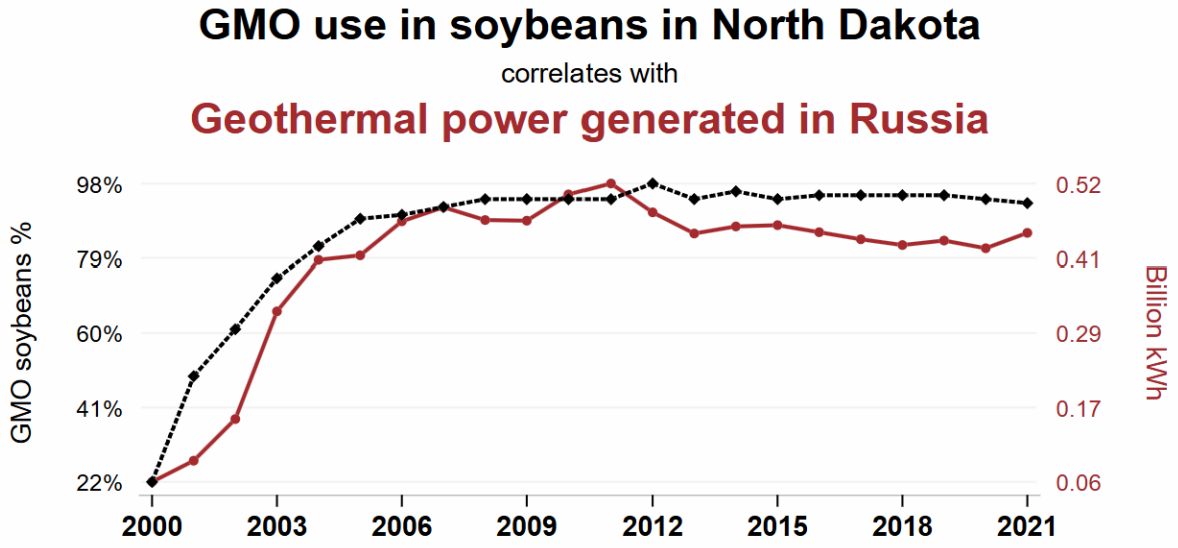
CIRCUMSTANCE 3

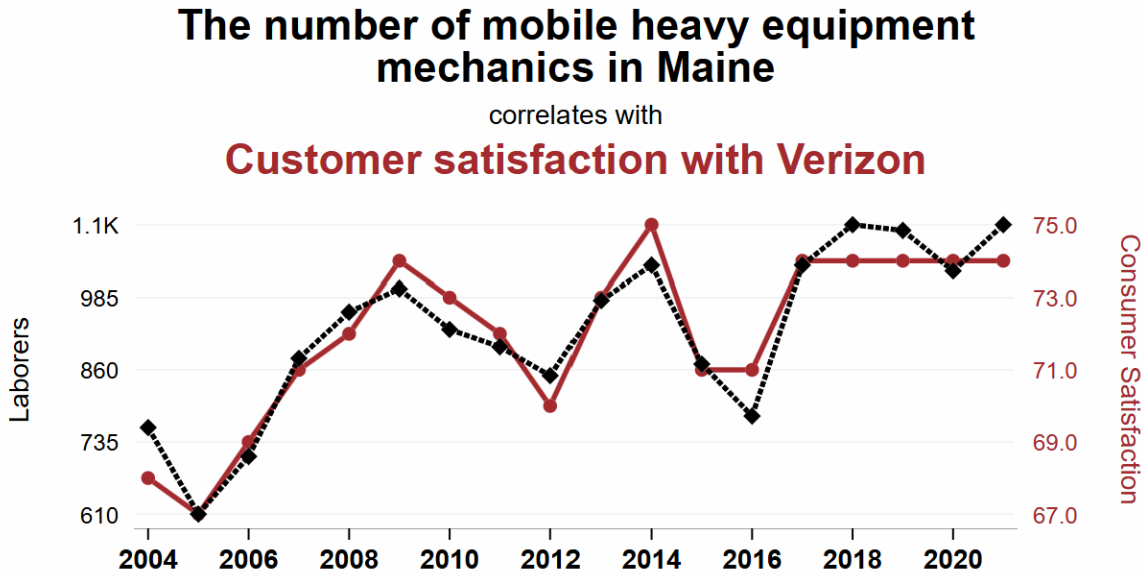
A confounded relationship: a hidden third factor influence both variables. This has a real structure behind it. A spurious one has none — and its only remedy is not to have gone looking.

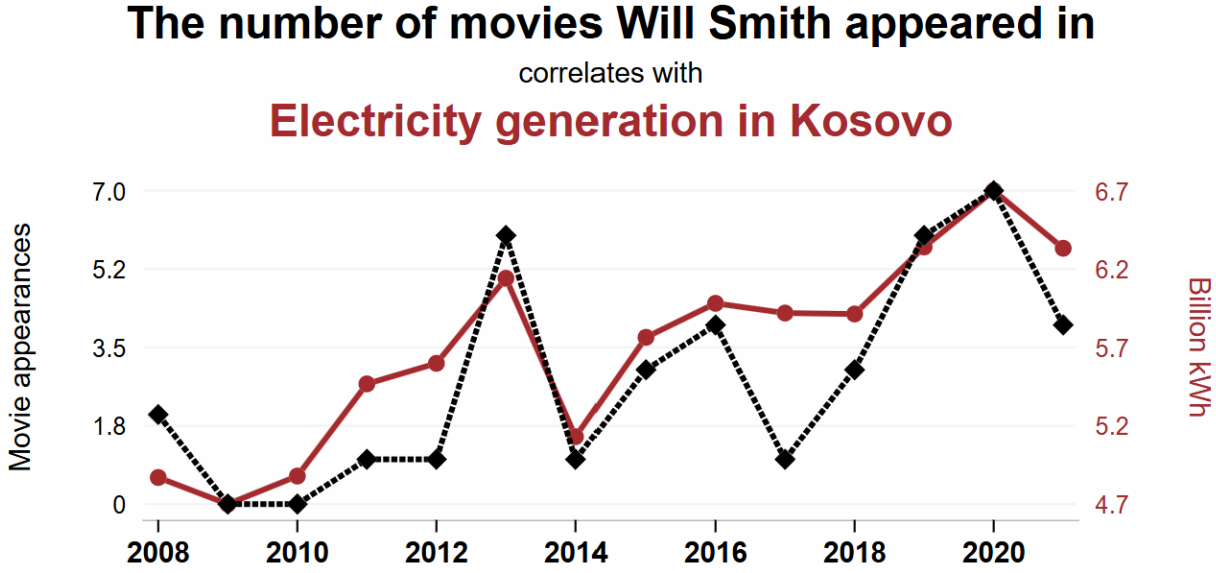
Spurious Correlation (Tương quan giả)

Source: https://www.tylervigen.com/spurious-correlations









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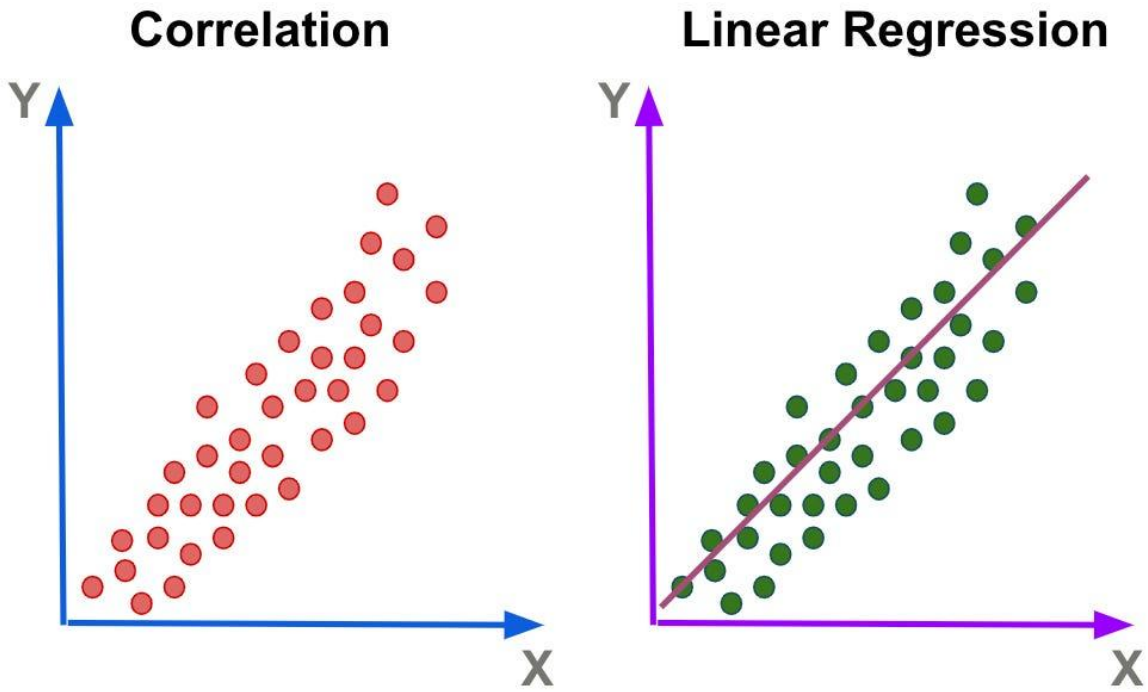
The three conditions, and the difference between predicting and intervening

PREDICTING ONE VARIABLE FROM ANOTHER

Linear Regression

Independent Variable (X): The input or predictor variable used to make the prediction.
Dependent Variable (Y): The target or response variable you want to predict.

- Plot and examine the scatter diagram
- If a linear relation emerges, try to fit a line through the data points
- The calculation of r will confirm the strength of this relation



$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$$

Slope $$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}} = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sum (x - \bar{x})^2}$$

Intercept $$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

Practice 3 — Three Lines Drawn by Hand

Three participants at the board · marker on the projected scatter

Three participants each draw the line that fits the cloud best. The three lines are then compared.

TIME

5 minutes

QUANTITY

Three lines, three people, one cloud

CONSTRAINT

No measuring, no calculation — draw the line the eye proposes.

The three lines stay on the board until the fitted line is computed.

Which of the three is correct?

Residuals and the Least-Squares Criterion

Residuals (e)

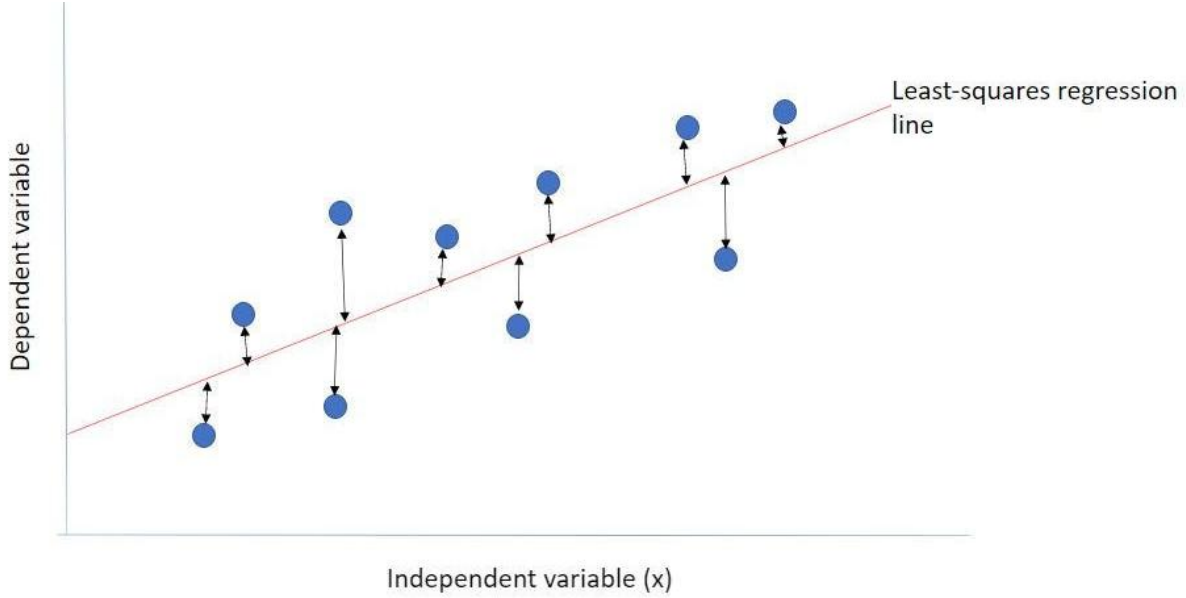
The residual is the vertical distance from a point to the line. Residuals Plot is a graph showing independent or predicted values on the horizontal axis and residuals on the vertical axis.

- Random Pattern: Points form an even, random cloud scattered around the horizontal zero line. This shows that the linear model fits well and errors are unbiased.

Why Sum of Squares (SS)

The line is chosen to minimize the sum of the squared residuals — for three reasons:

- Without squaring, positives and negatives cancel
- Squaring penalizes large errors disproportionately
- The minimization has a closed-form solution



$$SS_{(residuals)} = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

Fitting the Line in Excel

Three routes to the same two numbers, and three cells of twenty

Route	Returns
=SLOPE(Y,X) and =INTERCEPT(Y,X)	the two coefficients
Trendline > Display Equation, R-squared	the equation printed on the figure
Data Analysis > Regression	a table of about twenty numbers
Read today: slope ___, intercept ___, R^2 ___	the other seventeen are Sessions 9 and 12

Practice 4 — One Sentence, Written Individually

On paper · collected and read aloud

Write one sentence in Vietnamese stating what the slope of ROA on log(total assets) means.

TIME

2 minutes

QUANTITY

One sentence, no more

CONSTRAINT

The sentence must carry the units of both variables.

Four or five sentences are read aloud. One of them is copied onto the board and left there.

Three Kinds of Statement

Sentence	Status
Total assets rise by 1%, so ROA rises by ___ points	a causal verb, and one firm changing — not supported
A firm 1% larger has ROA higher by ___ points	supported
…higher by ___% rather than ___ percentage points	a unit error

The causal claim enters through the grammar, and most of the room wrote it without noticing.

The Standard Sentence

"An observation with __ [Unit of X] higher [Variable X]
has [Variable Y] higher/lower by __ [Unit of Y]“

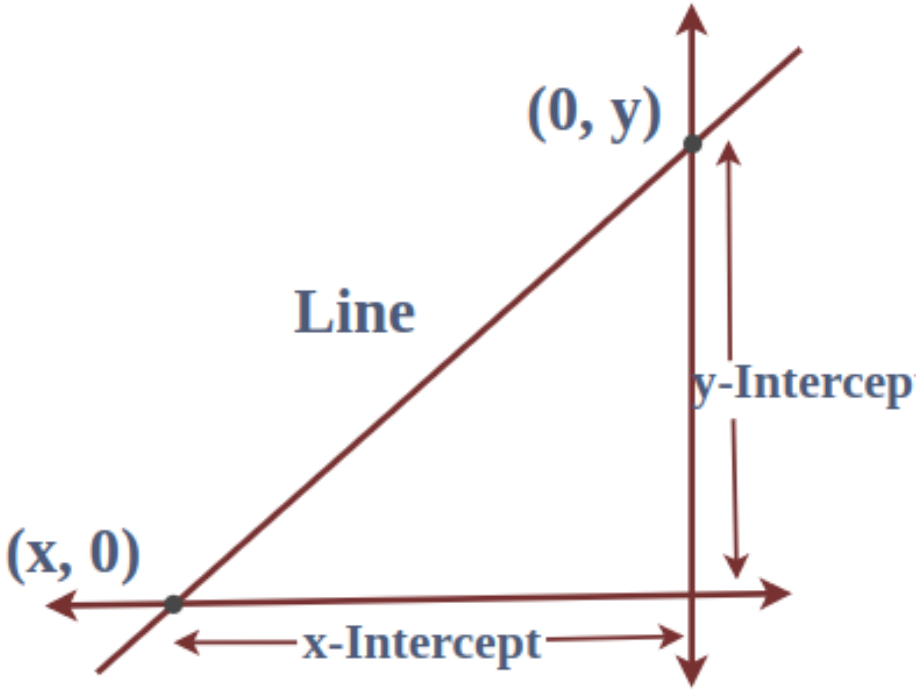
The sentence compares two different variables at one point in time. It does not describe one variable changing.

The Intercept, and the Range of the Line

WHERE THE LINE HAS STANDING

The intercept is the fitted value at X = 0 or Y = 0. No firm in this dataset lies near zero, so it is a number required by the algebra of the line rather than an economic quantity.

- Outside the observed range the line is continued on paper
- No observation lies beneath it there
- This limitation has a name

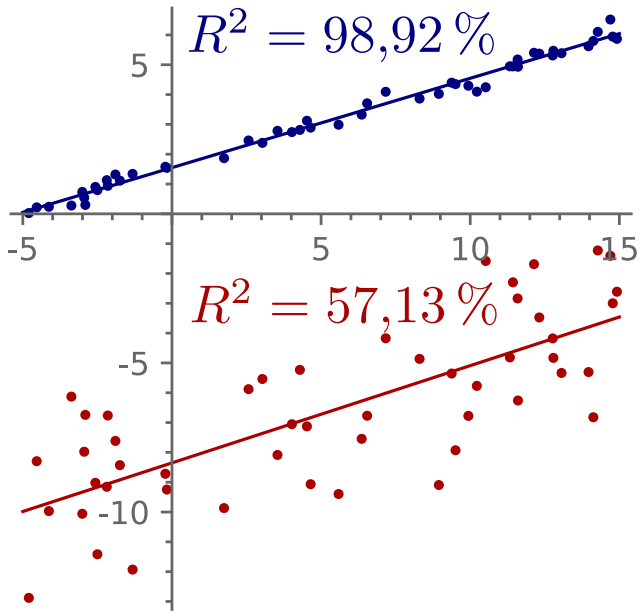


PREDICTING ONE VARIABLE FROM ANOTHER

The Coefficient of Determination

In a regression with one explanatory variable, $R^2 = r^2$

Statement	Status
R^2 = ___ on this model	the line accounts for ___% of the variation in ROA
A low R^2 means the model is useless	false — behavioural models report 0.1 to 0.3 and are published
A high R^2 means the model is correct	false — two trending series give a high R^2 and no relationship



Three Residual Patterns

Scattered about zero

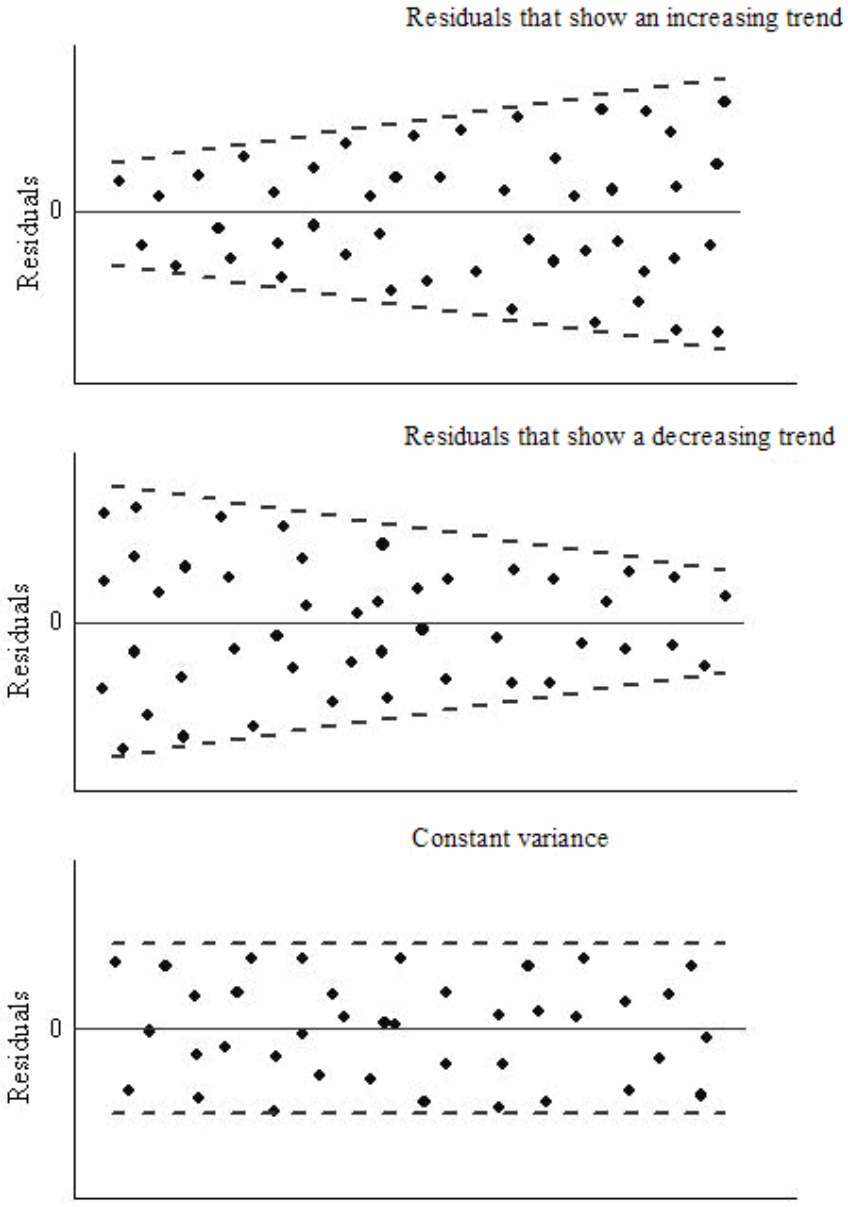
A straight line is an adequate form here.

A U shape

The relationship is curved; the straight line has the wrong form.

A widening funnel

The spread of Y increases with X.



PREDICTING ONE VARIABLE FROM ANOTHER

Practice 5 — The Slope Is One Draw

Excel · the Session 6 simulation workbook, new sheet

Draw two columns instead of one and replace AVERAGE by SLOPE. Press F9. Then copy down to 400 rows and plot the histogram of the 400 slopes.

TIME

4 minutes

QUANTITY

400 samples of n = 40, one slope each

CONSTRAINT

Mark the slope read at minute 60 on the histogram. It will not sit at the centre.

The mean of the 400 slopes, and their standard deviation.

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Without a Causal Verb

No causal verb was used since the beginning of this lecture

Firms in this industry are profitable more often — not this industry makes firms profitable.

Firms with higher debt have lower ROA — not debt reduces ROA.

The gap between what was measured and what was heard is the subject of the remaining

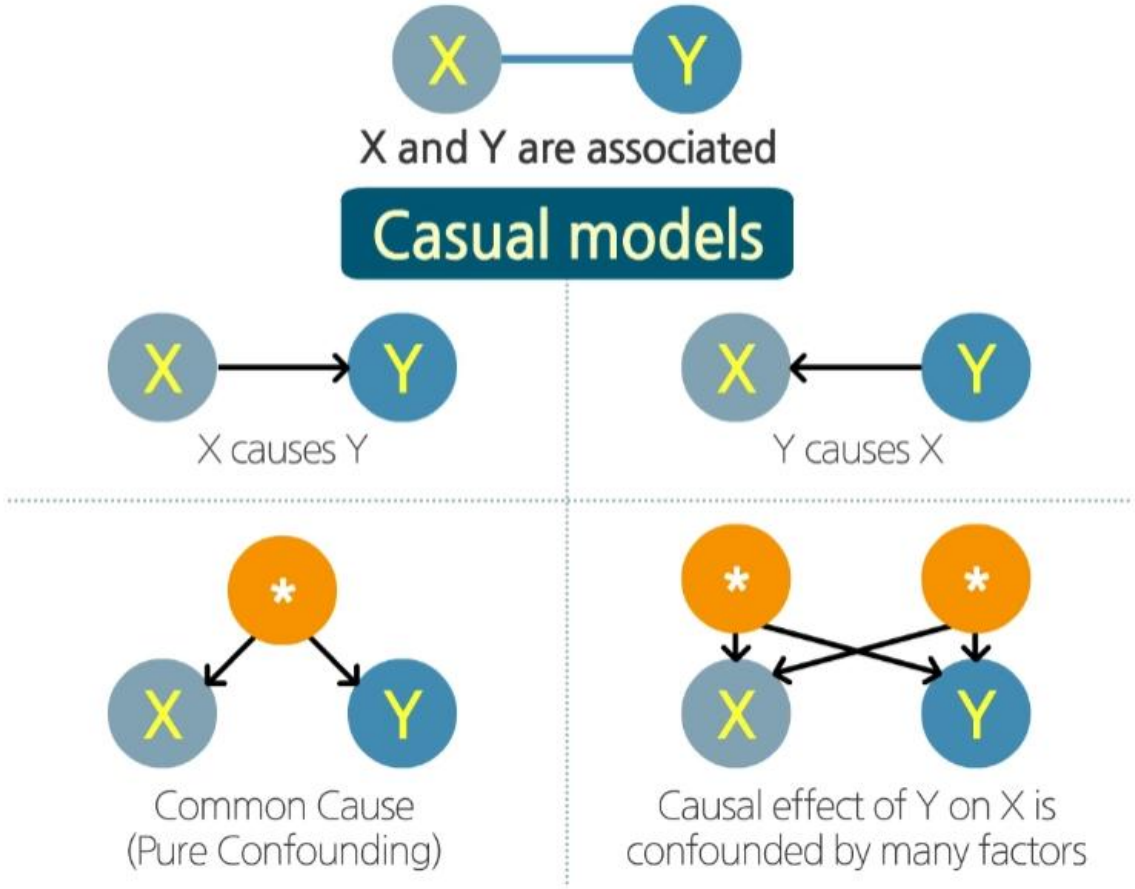
INTERPRETING CAUSALITY

THE SAME FOUR NUMBERS

Three causal worlds, one set of coefficients

Three structures are consistent with the same observed association between X and Y, and they produce identical r, identical slope, identical R^2 and identical residual plots.

- Direct effect — X → Y
- Confounding — X ← Z → Y, with no arrow between X and Y
- Reverse causation — X ← Y



No computation performed on a cross-section distinguishes them.

Confounding, Shown on a Contingency Table

Size class	Industry ___	All others	Difference
whole market	___%	___%	___ points
small firms	___%	___%	___ points
medium firms	___%	___%	___ points
large firms	___%	___%	___ points

A third column was doing the work, and nothing in the first table disclosed it.

Reverse Causation: Two Accounts of One Dataset

	ROA on size	Size on ROA
Slope	___	___
r	___	the same
R^2	___	the same
Size → performance	scale economies, fixed costs spread, bargaining power	
Performance → size	profitable firms accumulate, reinvest, borrow, and grow	

The two accounts imply opposite policies, and a cross-section stands still between them.

INTERPRETING CAUSALITY

Self-Selection

A special case of confounding, and the one most often met in a thesis

OBSERVED

Members of the loyalty scheme spend 40% more — or customers who already spent heavily are the ones who enrolled.

OBSERVED

Firms listed on HOSE report higher ROA than firms on HNX — or only firms already performing well meet the listing requirements.

THE COMMON FEATURE

The unit chooses its own group. Whatever determined that choice is liable to be the same thing that determines the outcome.

What the Arithmetic Cannot Separate

THREE STRUCTURES

Direct effect, confounding and reverse causation each correspond to something true about the world: a third variable exists, or the arrow points the other way.

- And all three produce the same results, but way different interpretation

A FOURTH CASE

Spurious correlation has no mechanism at all

Two columns happen to move together. This is what a search through a correlation matrix finds, and it is the only one of the four where the error lies in how the relationship was found.

- Not in the world — in the searching

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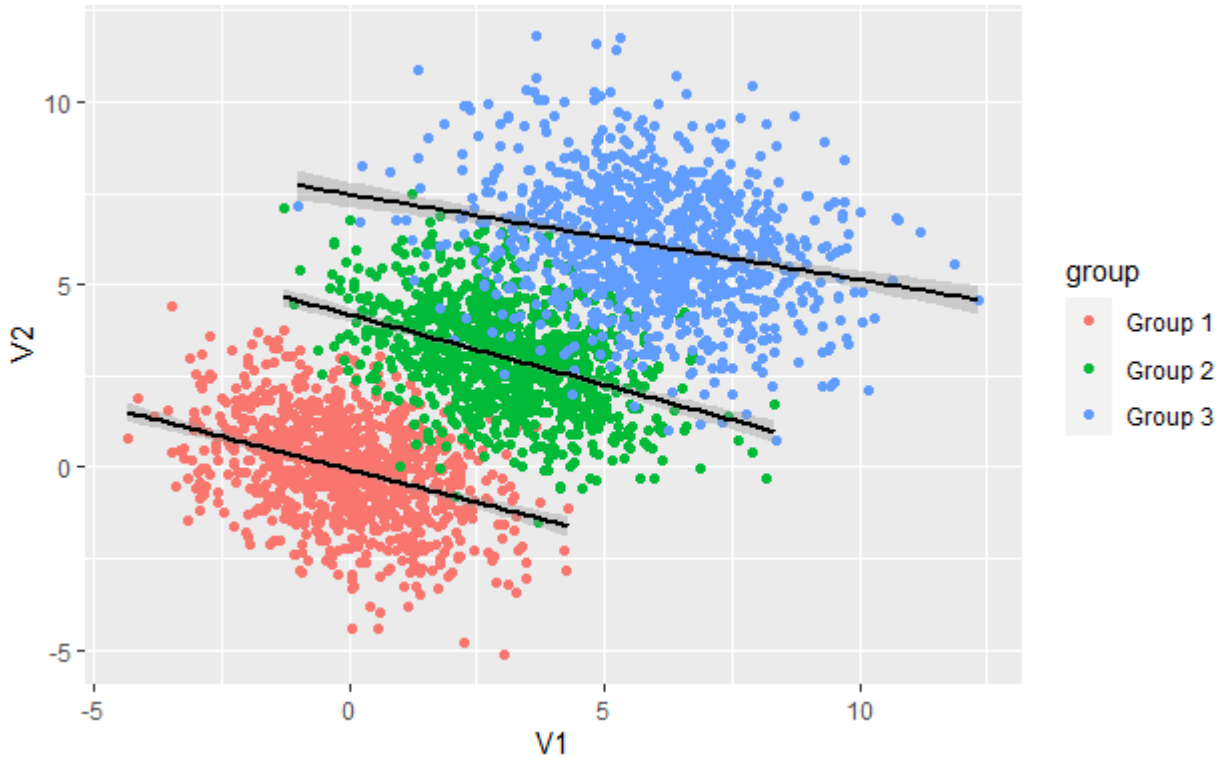
Why the Reversal Is Possible

THE PARADOX

Within each industry the direction falls; combined it rises

A statistical phenomenon where a trend or relationship appearing within separate groups of data disappears or reverses completely when those groups are combined

- **Confounding Variables**: A hidden third factor (or lurking variable) influences both the grouping and the outcome.
- **Unequal Group Sizes**: Aggregated data gets skewed when larger groups or groups with extreme baseline rates dominate the combined total.
- **Causal Misinterpretation**: Looking only at the surface-level grand total can lead to false conclusions about cause and effect



Neither number is incorrect. They answer two different questions.

SIMPSON'S PARADOX

Leverage and Profitability, Market-Wide

Firms split at the median debt ratio · row percentages

	Profitable
High leverage	___%
Low leverage	___%
Market-wide, the more leveraged group is profitable more often	

SIMPSON'S PARADOX

So which answer is correct?

THE AGGREGATE ANSWERS

Among all listed firms

How does profitability compare across leverage levels? The combined table answers this, and it says the more leveraged group is profitable more often.

- True, and rarely the question intended

THE WITHIN-GROUP ANSWERS

For a firm in a given industry

How does profitability compare across leverage levels? The eight tables answer this, and they say the opposite. This is the question almost always intended.

- Session 12 admits industry as dummy variables
- The model does not decide whether industry belongs there

Whether to aggregate or disaggregate is not a statistical choice. It comes from understanding the industry.

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From the scatter diagram to r — and the four things r cannot see

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PREDICTING ONE VARIABLE FROM ANOTHER

The fitted line, the slope in its correct units, and what the slope is not

04

INTERPRETING CAUSALITY

Three causal worlds that produce identical coefficients

05

SIMPSON'S PARADOX

An aggregate comparison that reverses inside every single group

06

FROM NUMBER TO CLAIM

The three conditions, and the difference between predicting and intervening

Three Minimum Conditions for a Causal Claim

#	Condition	Countered by	Established by
1	X and Y are associated	—	a table, r, a slope — the data answers this
2	X precedes Y	reverse causation	design; data with a time dimension
3	Alternatives are excluded	confounding, self-selection, Simpson	subject knowledge; controlling for variables

Condition 1 is the only one a computation settles. Today: 72 minutes on it, 58 on the other two.

FROM NUMBER TO CLAIM

Prediction and Intervention

PREDICTION

Which firms are likely to default next year?

Causation is not required. Any variable that predicts well may be used, whether or not it acts on anything.

- The name given to that step — prediction of one variable from another is limited to “Prediction” only

INTERVENTION

If the debt ratio is reduced, will ROA rise?

Causation is required. This is the question for which the preceding content were necessary.

- Settle which one you are doing before the analysis, not after

What we done today only answer the question about Prediction!